

Complex Analysis Qualifying Exam Review

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Chapter 1

Basic calculus in the complex domain

Proposition 1.1 (Cauchy-Riemann Equations). If $f : \Omega \rightarrow \mathbb{C}$ is holomorphic, then $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist and are continuous on Ω , and

$$\frac{\partial f}{\partial x} = \frac{1}{i} \frac{\partial f}{\partial y}$$

If we write $f = u + iv$, then the above is equivalent to

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases}$$

Proposition 1.2 (Converse of Cauchy-Riemann). If $f : \Omega \rightarrow \mathbb{C}$ is C^1 and satisfies Cauchy-Riemann equations, then f is holomorphic.

Proposition 1.3. Let $f : \Omega \rightarrow \mathbb{C}$ be holomorphic, where Ω is open. Let $p \in \Omega$, assume $f'(p) \neq 0$. Then there exists a neighborhood O of p and a neighborhood U of $q = f(p)$ such that $f : O \rightarrow U$ is bijective, where the inverse $g = f^{-1} : U \rightarrow O$ is holomorphic. For $z \in O, w = f(z)$, we have

$$g'(w) = \frac{1}{f'(z)}$$

Proposition 1.4. Suppose $\Omega \subset \mathbb{C}$ is convex. Assume f is holomorphic in Ω and there exists $a \in \mathbb{C}$ such that

$$\operatorname{Re} a f' > 0 \text{ on } \Omega$$

then f is one-to-one onto its image $f(\Omega)$.

Chapter 2

Cauchy integral theorem

Theorem 2.1. If $f \in C^1(\overline{\Omega})$ is holomorphic on Ω , then

$$\int_{\partial\Omega} f(z)dz = 0$$

Theorem 2.2. If $f \in C^1(\overline{\Omega})$ is holomorphic on Ω and $z_0 \in \Omega$, then

$$f(z_0) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(z)}{z - z_0} dz$$

One can change $\partial\Omega$ to a simple closed curve γ completely contained in Ω .

Theorem 2.3 (holomorphic implies analytic). If $f \in C^1(\overline{\Omega})$ is holomorphic, then for $z \in D_r(z_0) \subset \Omega$, then $f(z)$ has the convergent power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where

$$a_n = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = \frac{f^{(n)}(z_0)}{n!}$$

Proposition 2.1. If f is holomorphic on an open set $\Omega \subset \mathbb{C}$, then

$$f \in C^\infty(\Omega)$$

Proposition 2.2. Let $\Omega \subset \mathbb{C}$ be an open set and let $f_\nu : \Omega \rightarrow \mathbb{C}$ be holomorphic. Assume $f_\nu \rightarrow f$ locally uniformly, then $f : \Omega \rightarrow \mathbb{C}$ is holomorphic (and $f'_\nu \rightarrow f'$ locally uniformly on Ω).

Proposition 2.3 (maximum modulus principle). Let $\Omega \subset \mathbb{C}$ be connected, open. If f is holomorphic in Ω , then for $z_0 \in \Omega$,

$$\text{if } |f(z_0)| = \sup_{z \in \Omega} |f(z)|, \text{ then } f \text{ is constant on } \Omega$$

Moreover, if Ω is bounded and $f \in C(\overline{\Omega})$, then

$$\sup_{z \in \overline{\Omega}} |f(z)| = \sup_{z \in \partial\Omega} |f(z)|$$

(In this case, f must attain a maximum in $\overline{\Omega}$). This basically states that either f is constant or f attains maximum on the boundary.

Theorem 2.4 (Liouville's theorem). If $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic and bounded, then f is constant.

A consequence of Liouville's theorem is the fundamental theorem of algebra:

Proposition 2.4. If $p(z)$ is a polynomial of degree $n \geq 1$, then $p(z)$ has at least one root. (Proof: consider $1/p(z)$).

There are several properties that also hold for harmonic functions.

Proposition 2.5 (mean value property of harmonic functions). If $u \in C^2(\Omega)$ is harmonic, $z_0 \in \Omega$, and $\overline{D_r(z_0)} \subset \Omega$, then

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta}) d\theta$$

Proposition 2.6. Let $\Omega \subset \mathbb{C}$ be connected and open. If $u : \Omega \rightarrow \mathbb{R}$ is harmonic in Ω , then for $z_0 \in \Omega$,

$$u(z_0) = \sup_{z \in \Omega} u(z) \Rightarrow u \text{ is constant on } \Omega$$

Moreover, if Ω is bounded, $u \in C(\overline{\Omega})$, then

$$\sup_{z \in \overline{\Omega}} u(z) = \sup_{z \in \partial\Omega} u(z)$$

Proposition 2.7 (Liouville for harmonic functions). If $u \in C^2(\mathbb{C})$ is bounded and harmonic on all of $\mathbb{C}$, then u is constant.

Proposition 2.8. Let $\Omega \subset \mathbb{C}$ be simply connected domain. Then each harmonic function $u \in C^2(\Omega)$ has a harmonic conjugate.

Next is a converse of Cauchy's integral theorem.

Theorem 2.5 (Morera's theorem). Let $g : \Omega \rightarrow \mathbb{C}$ be continuous, and

$$\int_{\gamma} g(z) dz = 0$$

for any $\gamma = \partial R$, where $R \subset \Omega$ is a rectangle. Then g is holomorphic.

Proposition 2.9 (Schwarz reflection principle). Assume $f : \Omega^+ \cup L \rightarrow \mathbb{C}$ is continuous, holomorphic in Ω^+ , and real-valued on L . Define $g : \Omega \rightarrow \mathbb{C}$ by

$$g(z) = f(z), z \in \Omega^+ \cup L, \overline{f(\bar{z})}, z \in \Omega^-$$

Then g is holomorphic on Ω .

Proposition 2.10 (Goursat's theorem). If $f : \Omega \rightarrow \mathbb{C}$ is complex-differentiable at each point of Ω , then f is holomorphic.

Proposition 2.11 (analytic continuation). Let $\Omega \subset \mathbb{C}$ be open and connected, and let $f : \Omega \rightarrow \mathbb{C}$ be holomorphic. If $f = 0$ on a set $S \subset \Omega$ and S has an accumulation point $p \in \Omega$, then $f = 0$ on Ω .

In other words, if $f = g$ on a set $S \subset \Omega$, where S contains an accumulation point, then $f = g$ on Ω .

Definition 2.1 (singularities). There are three types of singularities: p is said to be the following for $f : U \rightarrow \mathbb{C}$,

1. Removable: if there exists $\tilde{f}$ on U such that $\tilde{f} = f$ on $U \setminus p$.
2. Pole: if $|f(z)| \rightarrow \infty$ as $z \rightarrow p$. Equivalently, p is a removable singularity for $1/f$.
3. Essential singularity: not removable, not pole, an isolated singularity. $f(z) = e^{1/z}$.

Proposition 2.12. If $p \in \Omega$ and f is holomorphic on $\Omega \setminus p$ and bounded, then p is a removable singularity.

Proposition 2.13. If f is holomorphic on $\Omega \setminus p$ with a pole at p , then there exists $k \in \mathbb{Z}^+$ such that

$$f(z) = \frac{F(z)}{(z-p)^k}$$

on $\Omega \setminus p$, where F is holomorphic on Ω and $F(p) \neq 0$.

Proposition 2.14 (Casorati-Weierstrass). Suppose $f : \Omega \setminus p \rightarrow \mathbb{C}$ has an essential singularity at p . Then for any neighborhood of U of p in Ω , the image of $U \setminus p$ is dense in $\mathbb{C}$.

Proposition 2.15. If $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic and $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, then $f(z)$ is a polynomial.

Proposition 2.16. Let $f : \mathcal{A} \rightarrow \mathbb{C}$ be holomorphic, where the annulus $\mathcal{A} = \{z : r_0 \leq |z - z_0| \leq r_1\}$, then f is given by

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, z \in \mathcal{A} \quad (\text{Laurent series})$$

and

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta, n \in \mathbb{Z}$$

where γ is any counterclockwise circle centered at z_0 , of radius $r \in (r_0, r_1)$.

How does the Laurent series expansion help us?

Proposition 2.17. Suppose f is holomorphic on $D_R(z_0) \setminus z_0$, with the above Laurent series. f has a pole at z_0 iff $a_n = 0$ for all but finitely many $n < 0$; f has an essential singularity iff $a_n \neq 0$ for infinitely many n .

Chapter 3

Fourier analysis and complex function theory

Proposition 3.1. Let $f \in C(S^1)$, if the Fourier coefficients $\hat{f}(k)$ form a summable series, i.e., if

$$\sum_{k=-\infty}^{\infty} |\hat{f}(k)| < \infty$$

then for each $\theta \in S^1$,

$$f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ik\theta}$$

Proposition 3.2. If f is such that $\sum |\hat{f}(k)| < \infty$, then

$$\sum |\hat{f}(k)|^2 = \frac{1}{2\pi} \int_{S^1} |f(\theta)|^2 d\theta$$

More generally, if also g is such that $\sum |\hat{g}(k)| < \infty$,

$$\sum \hat{f}(k) \overline{\hat{g}(k)} = \frac{1}{2\pi} \int_{S^1} f(\theta) \overline{g(\theta)} d\theta$$

Proposition 3.3. If f is a continuous, piecewise C^1 function on S^1 , then $\sum |\hat{f}(k)| < \infty$.

Proposition 3.4. If f is Riemann integrable on S^1 , then $S_N f \rightarrow f$ in L^2 , and

$$\sum |\hat{f}(k)|^2 = \|f\|_{L^2}^2$$

Proposition 3.5 (Dirichlet boundary problem). If $D \subset \mathbb{C}$ is an open disk and given $f \in C(\partial D)$, there exists a unique $u \in C(\overline{D}) \cap C^2(D)$ satisfying

$$\Delta u = 0 \text{ on } D, u|_{\partial D} = f$$

Proposition 3.6. Assume $u : \Omega^+ \cup L \rightarrow \mathbb{C}$ is continuous, harmonic on Ω^+ , and $u = 0$ on L . Define $v : \Omega \rightarrow \mathbb{C}$ as

$$\begin{cases} v(z) = u(z), z \in \Omega^+ \cup L \\ -u(\bar{z}), z \in \Omega^- \end{cases}$$

then v is harmonic on Ω .

Proposition 3.7. If f is integrable on $\mathbb{R}$, then $\hat{f}$ is continuous on $\mathbb{R}$.

The next proposition gives criterion for the Fourier inversion formula.

Proposition 3.8. Assume f is bounded and continuous on $\mathbb{R}$, and $f, \hat{f}$ are integrable on $\mathbb{R}$. Then the Fourier inversion formula holds: for all $x \in \mathbb{R}$,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{ix\xi} d\xi$$

We define

$$S_R f(x) = \frac{1}{\sqrt{2\pi}} \int_{-R}^R \hat{f}(\xi) e^{ix\xi} d\xi$$

Proposition 3.9. If $f \in L^2(\mathbb{R})$, then as $R \rightarrow \infty$,

$$S_R f \rightarrow f \text{ in } L^2$$

Proposition 3.10 (uniqueness). If $f \in L^1(\mathbb{R})$ and $\hat{f} = 0$, then

$$f = 0$$

Definition 3.1. Let $f : \mathbb{R}^+ \rightarrow \mathbb{C}$ that is integrable on $[0, R]$, for all $R < \infty$, such that

$$\int_0^\infty |f(t)| e^{-at} dt < \infty$$

for all $a > A$, for some $A \in (-\infty, \infty)$. Then we define the Laplace transform of f

$$\mathcal{L}f(s) = \int_0^\infty f(t) e^{-st} dt, \operatorname{Re} s > A$$

Chapter 4

Residue calculus, the argument principle, and two very special functions

Definition 4.1. If f has the Laurent series in disk D_j around $p_j \in \Omega$,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(p_j)(z - p_j)^n$$

where the coefficient $a_{-1}(p_j)$ is called the residue of f at p_j , denoted as $\text{Res}_{p_j} f$. We have

$$\text{Res}_{p_j} f = \frac{1}{2\pi i} \int_{\partial D_j} f(z) dz$$

Proposition 4.1 (Residue formula). Let f be a meromorphic function on Ω , then for a simple closed curve C , we have

$$\int_C f(z) dz = 2\pi i \sum_{j=1}^k \text{Res}(f, z_j)$$

where the isolated singularities z_j 's are contained in C .

Proposition 4.2 (Residue). If f has a simple pole at z_0 , then

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

If f has a pole of order n at z_0 , then

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}} ((z - z_0)^n f(z))$$

If f has an essential singularity at z_0 , then we write f as Laurent series around z_0 ,

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where

$$\text{Res}(f, z_0) = c_{-1}$$

Proposition 4.3. Let $f \in C^2(\bar{\Omega})$ be holomorphic in Ω , nowhere zero on $\partial\Omega$, then let $\nu(f, \Omega)$ be the number of zeros of f in Ω , counted with multiplicity,

$$\nu(f, \Omega) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz$$

Proposition 4.4. Let C_j denote the connected component of $\partial\Omega$, and $\gamma_j = f(C_j)$. Then the winding number of γ_j around 0 satisfies

$$\frac{1}{2\pi i} \int_{\gamma_j} \frac{1}{z} dz = \frac{1}{2\pi i} \int_{C_j} \frac{f'(z)}{f(z)} dz$$

Definition 4.2 (winding number). Let γ be a curve given by

$$\gamma(t) = r(t)e^{i\theta(t)}$$

where $r(t), \theta(t)$ both continuous, piecewise C^1 , and $r(t) > 0$. The **winding number** of γ about 0 is given by

$$n(\gamma, 0) = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z} dz$$

Proposition 4.5. If γ_0, γ_1 are smoothly homotopic in $\mathbb{C} \setminus 0$, then

$$n(\gamma_0, 0) = n(\gamma_1, 0)$$

Combining all the above,

$$\nu(f, \Omega) = \sum_j n(\gamma_j, 0)$$

the number of zeros of f , counted with multiplicity, is equal to the sum of the winding numbers of $f(C_j)$ about 0.

Proposition 4.6 (Argument principle for meromorphic functions). If f is meromorphic on a bounded domain Ω and C^1 on a neighborhood of $\partial\Omega$. Then the number of zeros of f minus the number of poles, counted with multiplicity is equal to the sum of winding numbers of $f(C_j)$ around 0, where C_j are the connected components of $\partial\Omega$.

Theorem 4.1 (Rouche's theorem). Let $f, g \in C^1(\bar{\Omega})$ be holomorphic in Ω , and nowhere zero on $\partial\Omega$. Assume

$$|f(z) - g(z)| < |f(z)|, \text{ for all } z \in \partial\Omega$$

then

$$\nu(f, \Omega) = \nu(g, \Omega)$$

Proposition 4.7 (Open mapping theorem). If $\Omega \subset \mathbb{C}$ is open and connected and $f : \Omega \rightarrow \mathbb{C}$ is holomorphic and non-constant, then f maps open sets to open sets.

Definition 4.3 (Gamma function). The gamma function Γ is given by

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt, \operatorname{Re} z > 0$$

Definition 4.4 (Riemann zeta function). The zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \operatorname{Re} s > 1$$

Proposition 4.8. For $z \in \mathbb{C} \setminus \mathbb{Z}$, we have

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

Proposition 4.9. $\Gamma(z)$ has no zeros, so $1/\Gamma(z)$ is an entire function.

Proposition 4.10. For $\operatorname{Re} z > 0$, we have

$$\Gamma(z) = \lim_{n \rightarrow \infty} n^z \frac{1 \cdot 2 \cdots n}{z(z+1) \cdots (z+n)}$$

Proposition 4.11 (Euler's product formula). Let $\{p_j : j \geq 1\}$ denote the set of prime numbers, then for $\operatorname{Re} s > 1$,

$$\zeta(s) = \prod_{j=1}^{\infty} \frac{1}{1 - p_j^{-s}}$$

Proposition 4.12. The function $\zeta(s)$ extends to a meromorphic function on $\mathbb{C}$, with one simple pole at $s = 1$. Moreover,

$$\zeta(s) - \frac{1}{s-1}$$

extends to an entire function of s .

Chapter 5

Conformal maps and geometrical aspects of complex function theory

Definition 5.1 (conformal). We say f is conformal at z_0 if γ_1, γ_2 are two smooth paths that intersect at z_0 , then the images $f(\gamma_j)$ intersect at $f(z_0)$ at the same angle as γ_1 and γ_2 .

Proposition 5.1. A smooth function f on Ω is holomorphic iff it is conformal and preserves orientation.

Theorem 5.1 (Riemann mapping theorem). If $\Omega \subset \mathbb{C}$ is a simply connected domain and $\Omega \neq \mathbb{C}$, then there exists a holomorphic diffeomorphism

$$F : \Omega \rightarrow D$$

where D is the unit disk. (A holomorphic diffeomorphism is a holomorphic function that has a holomorphic inverse).

The following is an improvement of Casorati-Weierstrass theorem.

Theorem 5.2 (Picard's theorem). Let p, q be distinct and

$$f : D \setminus \{0\} \rightarrow \mathbb{C} \setminus \{p, q\}$$

is holomorphic, then the singularity at 0 is either a pole or a removable singularity.

Proposition 5.2 (Schwarz Lemma). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic with $f(0) = 0$, then

1. $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
2. For some $z_0 \neq 0$, if $|f(z_0)| = |z_0|$, then f is a rotation.
3. We have $|f'(0)| \leq 1$, and if it attains equality, then it implies f is a rotation.